

Deep venous thrombosis

Blood clot within a deep vein → swelling, redness **A**, warmth, pain. Predisposed by Virchow triad (**SHE**):

- **S**tasis (eg, post-op, long drive/flight)
- **H**ypercoagulability (eg, defect in coagulation cascade proteins, such as factor V Leiden; oral contraceptive use; pregnancy)
- **E**ndothelial damage (exposed collagen triggers clotting cascade)

DVT of proximal deep veins of lower extremity (iliac, femoral, popliteal) → embolic source.

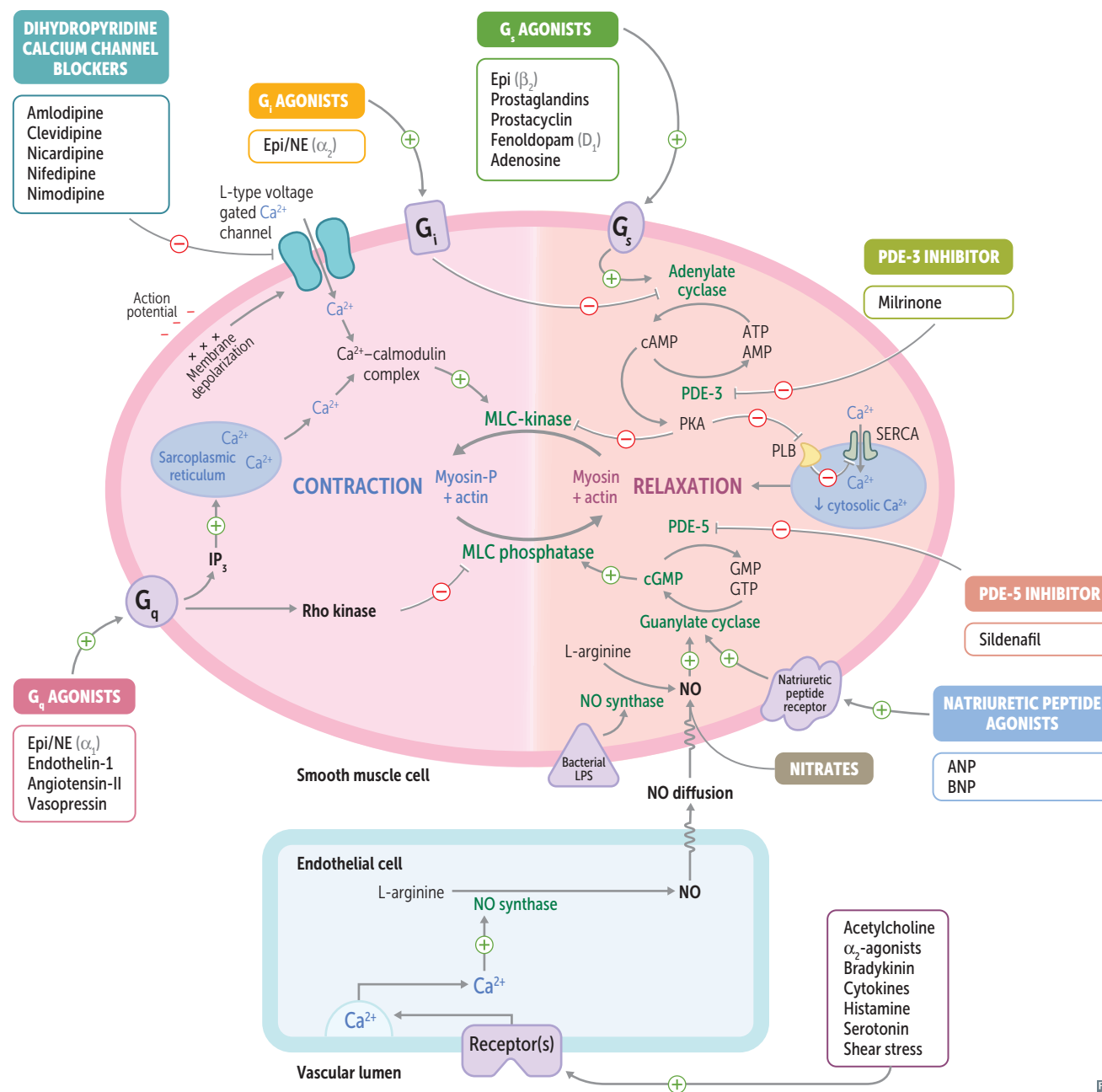
D-dimer test may be used clinically to rule out DVT if disease probability is low or moderate (high sensitivity, low specificity). Imaging test of choice is compression ultrasound with Doppler.

Use unfractionated heparin or low-molecular weight heparins (eg, enoxaparin) for prophylaxis and acute management. Use direct anticoagulants (eg, rivaroxaban, apixaban) for treatment and long-term prevention.

► CARDIOVASCULAR—PHARMACOLOGY**Hypertension treatment**

Primary (essential) hypertension	Thiazide diuretics, ACE inhibitors, angiotensin II receptor blockers (ARBs), Ca ²⁺ channel blockers.	
Hypertension with heart failure	ACE inhibitors/ARBs, β-blockers (compensated HF), aldosterone antagonists (mortality benefit), diuretics (improve symptoms and reduce hospitalization)	β-blockers must be used cautiously in decompensated HF and are contraindicated in cardiogenic shock.
Hypertension with diabetes mellitus	ACE inhibitors/ARBs, Ca ²⁺ channel blockers, thiazide diuretics, β-blockers.	ACE inhibitors/ARBs are protective against diabetic nephropathy. β-blockers can mask hypoglycemia symptoms.
Hypertension with asthma	ARBs, Ca ²⁺ channel blockers, thiazide diuretics, cardioselective β-blockers.	Avoid non-selective β-blockers to prevent β ₂ -receptor–induced bronchoconstriction. Avoid ACE inhibitors to prevent confusion between drug- or asthma-related cough.
Hypertension with pregnancy	H ydralazine, m ethyldopa, l abetalol, n ifedipine.	H ypertensive m oms l ove n ifedipine.
Hypertension with gout	ACE inhibitors, Ca ²⁺ channel blockers, ARBs (especially losartan).	Avoid loop and thiazide diuretics as these can cause gout flares.
Hypertension with osteoporosis	Thiazides preferred as monotherapy; combine with ACE inhibitors or ARBs if needed.	Thiazides preferred due to increased Ca ²⁺ reabsorption.
Hypertension with pheochromocytoma	Phentolamine, phenoxybenzamine (α-blockade first) prior to resection and β-blockade.	Avoid β-blockade first to mitigate unopposed α-mediated hypertensive crisis.

Cardiovascular agents and molecular targets

**Nitrates**

Nitroglycerin, isosorbide dinitrate, isosorbide mononitrate.

MECHANISMVasodilate by $\uparrow$ NO in vascular smooth muscle $\rightarrow \uparrow$ in cGMP and smooth muscle relaxation. Dilate veins $\gg$ arteries. $\downarrow$ preload.**CLINICAL USE**

Angina, ACS, pulmonary edema.

ADVERSE EFFECTSReflex tachycardia (treat with β -blockers), methemoglobinemia, hypotension, flushing, headache, "Monday disease" in industrial nitrate exposure: development of tolerance for the vasodilating action during the work week and loss of tolerance over the weekend $\rightarrow$ tachycardia, dizziness, headache upon reexposure. Contraindicated in right ventricular infarction, hypertrophic cardiomyopathy, and with concurrent PDE-5 inhibitor use.

Calcium channel blockers

Amlodipine, clevidipine, nicardipine, nifedipine, nimodipine (dihydropyridines, act on vascular smooth muscle); diltiazem, verapamil (nondihydropyridines, act on heart).

MECHANISM	Block voltage-dependent L-type calcium channels of cardiac and smooth muscle → ↓ muscle contractility. Vascular smooth muscle—amlodipine = nifedipine > diltiazem > verapamil. Heart—verapamil > diltiazem > amlodipine = nifedipine.
CLINICAL USE	Dihydropyridines (except nimodipine): hypertension, angina (including vasospastic type), Raynaud phenomenon. D ihydropyridine mainly d ilates arteries. Nimodipine: subarachnoid hemorrhage (prevents delayed ischemia). Nicardipine, clevidipine: hypertensive urgency or emergency. Nondihydropyridines: hypertension, angina, rate control in atrial fibrillation/flutter, prevention of nodal arrhythmias.
ADVERSE EFFECTS	Gingival hyperplasia. Dihydropyridine: peripheral edema, flushing, dizziness. Nondihydropyridine: cardiac depression, AV block, hyperprolactinemia (verapamil), constipation.

Hydralazine

MECHANISM	↑ cGMP → smooth muscle relaxation. Hydralazine vasodilates a rterioles > veins; a fterload reduction.
CLINICAL USE	Severe hypertension (particularly acute), HF (with organic nitrate). Safe to use during pregnancy. Frequently coadministered with a β-blocker to prevent reflex tachycardia.
ADVERSE EFFECTS	Compensatory tachycardia (contraindicated in angina/CAD), fluid retention, headache, angina, drug-induced lupus.

Hypertensive emergency

Treat with labetalol, clevidipine, fenoldopam, nicardipine, nitroprusside.

Nitroprusside	Short acting vasodilator (arteries = veins); ↑ cGMP via direct release of NO. Can cause cyanide toxicity (releases cyanide).
Fenoldopam	D opamine D ₁ receptor agonist—coronary, peripheral, renal, and splanchnic vasodilation. ↓ BP, ↑ natriuresis. Also used postoperatively as an antihypertensive. Can cause hypotension, tachycardia, flushing, headache, nausea.

Antianginal therapy

Goal is reduction of myocardial O₂ consumption (MVO₂) by ↓ 1 or more of the determinants of MVO₂: end-diastolic volume, BP, HR, contractility.

COMPONENT	NITRATES	β-BLOCKERS	NITRATES + β-BLOCKERS
End-diastolic volume	↓	No effect or ↑	No effect or ↓
Blood pressure	↓	↓	↓
Contractility	↑ (reflex response)	↓	Little/no effect
Heart rate	↑ (reflex response)	↓	No effect or ↓
Ejection time	↓	↑	Little/no effect
MVO ₂	↓	↓	↓↓

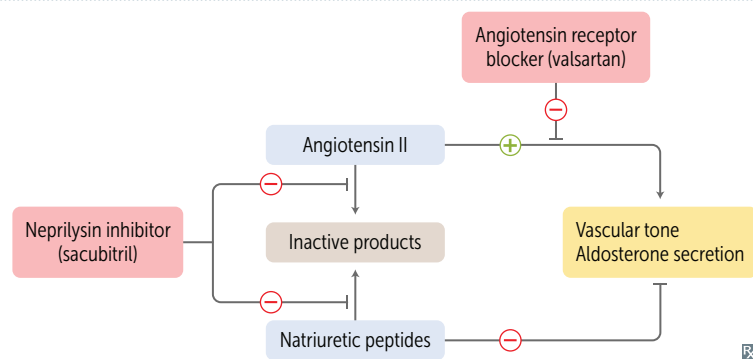
Nondihydropyridine calcium channel blockers (verapamil, diltiazem) are similar to β-blockers in effect.

Ranolazine

MECHANISM	Inhibits the late phase of inward sodium current thereby reducing diastolic wall tension and oxygen consumption. Does not affect heart rate or blood pressure.
CLINICAL USE	Refractory angina.
ADVERSE EFFECTS	Constipation, dizziness, headache, nausea, QT prolongation.

Sacubitril

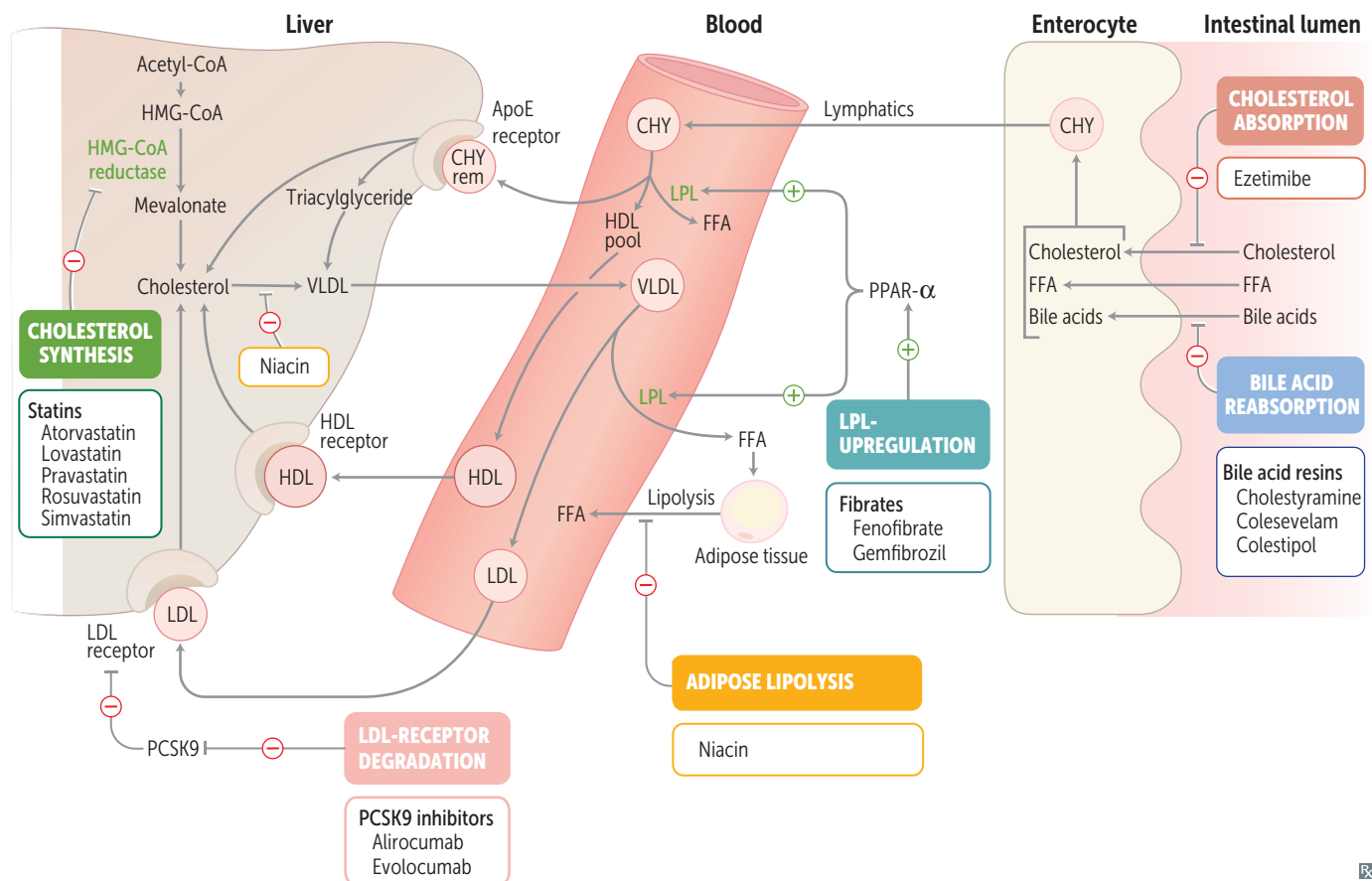
MECHANISM	A neprilysin inhibitor; prevents degradation of bradykinin, natriuretic peptides, angiotensin II, and substance P → ↑ vasodilation, ↓ ECF volume.
CLINICAL USE	Used in combination with valsartan (an ARB) to treat HFrEF.
ADVERSE EFFECTS	Hypotension, hyperkalemia (due to ARB component of HFrEF therapy), cough, dizziness; contraindicated with ACE inhibitors due to angioedema (both drugs ↑ bradykinin).

**Lipid-lowering agents**

DRUG	LDL	HDL	TRIGLYCERIDES	MECHANISM	ADVERSE EFFECTS
Statins Atorvastatin, lovastatin, pravastatin, rosuvastatin, simvastatin	↓↓↓	↑	↓	Inhibit HMG-CoA reductase → ↓ cholesterol synthesis; → ↓ intrahepatic cholesterol → ↑ LDL receptor recycling → ↑ LDL catabolism ↓ in mortality in patients with CAD	Hepatotoxicity (↑ LFTs), myopathy (especially when used with fibrates or niacin)
Bile acid resins Cholestyramine, colesevelam, colestipol	↓↓	↑ slightly	↑ slightly	Disrupt enterohepatic bile acid circulation → compensatory ↑ conversion of cholesterol to bile → ↓ intrahepatic cholesterol → ↑ LDL receptor recycling	GI upset, ↓ absorption of other drugs and fat- soluble vitamins
Ezetimibe	↓↓	↑/—	↓/—	Prevents cholesterol absorption at small intestine brush border	Rare ↑ LFTs, diarrhea

Lipid-lowering agents (continued)

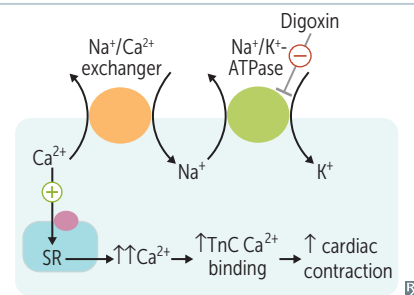
DRUG	LDL	HDL	TRIGLYCERIDES	MECHANISM	ADVERSE EFFECTS
Fibrates Fenofibrate, gemfibrozil	↓	↑	↓↓↓	Activate PPAR- α → upregulate LPL → ↑ TG clearance Activate PPAR- α → induce HDL synthesis	Myopathy (↑ risk with statins), cholesterol gallstones (via inhibition of cholesterol 7 α -hydroxylase)
Niacin	↓↓	↑↑	↓	Inhibits lipolysis (hormone-sensitive lipase) in adipose tissue; reduces hepatic VLDL synthesis	Flushed face (prostaglandin mediated; ↓ by NSAIDs or long-term use) Hyperglycemia Hyperuricemia
PCSK9 inhibitors Alirocumab, evolocumab	↓↓↓	↑	↓	Inactivation of LDL-receptor degradation → ↑ removal of LDL from bloodstream	Myalgias, delirium, dementia, other neurocognitive effects
Fish oil and marine omega-3 fatty acids	↑ slightly	↑ slightly	↓ at high doses	Believed to ↓ FFA delivery to liver, ↓ activity of TG-synthesizing enzymes, ↓ VLDL production, and inhibit synthesis of ApoB	Nausea, fishlike taste



Digoxin

MECHANISM

Direct inhibition of $\text{Na}^+/\text{K}^+-\text{ATPase}$.
 → indirect inhibition of $\text{Na}^+/\text{Ca}^{2+}$ exchanger.
 $\uparrow [\text{Ca}^{2+}]_i \rightarrow$ positive inotropy. Stimulates vagus nerve $\rightarrow \downarrow$ HR.



CLINICAL USE

HF ($\uparrow$ contractility); atrial fibrillation ($\downarrow$ conduction at AV node and depression of SA node).

ADVERSE EFFECTS

Cholinergic effects (nausea, vomiting, diarrhea), blurry **yellow** vision (“van **Glow**”), arrhythmias, atrial tachycardia with AV block.

Can lead to hyperkalemia, which indicates poor prognosis.

Factors predisposing to toxicity: renal failure ($\downarrow$ excretion), hypokalemia (permissive for digoxin binding at K^+ -binding site on $\text{Na}^+/\text{K}^+-\text{ATPase}$), drugs that displace digoxin from tissue-binding sites, and $\downarrow$ clearance (eg, verapamil, amiodarone, quinidine).

ANTIDOTE

Slowly normalize K^+ , cardiac pacer, anti-digoxin Fab fragments, Mg^{2+} .

Antiarrhythmics—sodium channel blockers (class I)

Slow or block conduction (especially in depolarized cells). $\downarrow$ slope of phase 0 depolarization.

$\uparrow$ action at **faster** HR. State dependent $\uparrow$ HR $\rightarrow$ shorter diastole, Na^+ channels spend less time in resting state (drugs dissociate during this state) $\rightarrow$ less time for drug to dissociate from receptor.

Effect most pronounced in **IC>IA>IB** due to relative binding strength. **Fast** taxi **CAB**.

Class IA

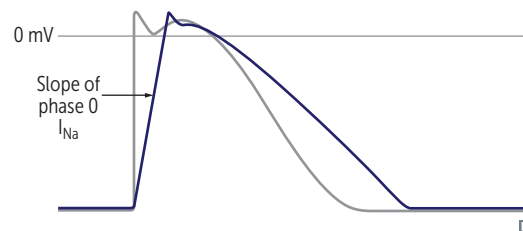
Quinidine, **pro**caïnamide, **diso**pyramide.

“The **queen** **pro**claims **Diso**’s **pyramid**.”

MECHANISM

Moderate Na^+ channel blockade.

$\uparrow$ AP duration, $\uparrow$ effective refractory period (ERP) in ventricular action potential, $\uparrow$ QT interval, some K^+ channel blocking effects.



CLINICAL USE

Both atrial and ventricular arrhythmias, especially reentrant and ectopic SVT and VT.

ADVERSE EFFECTS

Cinchonism (headache, tinnitus with quinidine), reversible SLE-like syndrome (procainamide), HF (disopyramide), thrombocytopenia, torsades de pointes due to $\uparrow$ QT interval.

Class IB

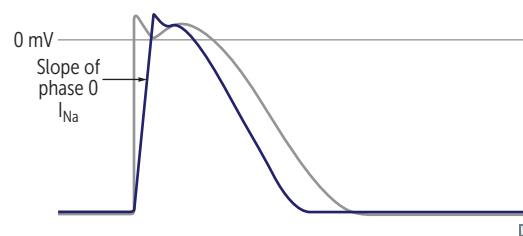
Lidocaine, **mex**iletine.

“T’**d** Buy **Liddy**’s **Mex**ican tacos.”

MECHANISM

Weak Na^+ channel blockade.

$\downarrow$ AP duration. Preferentially affect ischemic or depolarized Purkinje and ventricular tissue.



CLINICAL USE

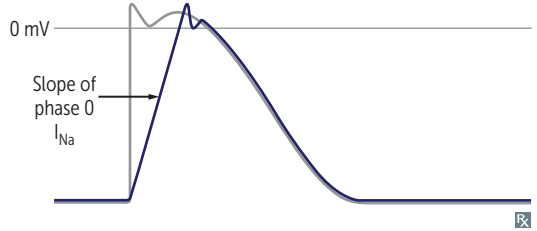
Acute ventricular arrhythmias (especially post-MI), digitalis-induced arrhythmias.

IB is **B**est post-MI.

ADVERSE EFFECTS

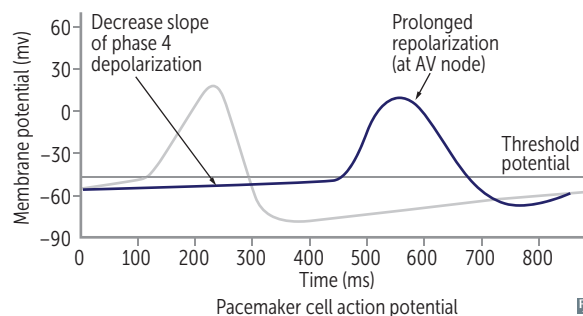
CNS stimulation/depression, cardiovascular depression.

Antiarrhythmics—sodium channel blockers (class I) (continued)

Class IC	Flecainide, propafenone. “Can I have fries, please?”	
MECHANISM	Strong Na ⁺ channel blockade. Significantly prolongs ERP in AV node and accessory bypass tracts. No effect on ERP in Purkinje and ventricular tissue. Minimal effect on AP duration.	
CLINICAL USE	SVTs, including atrial fibrillation. Only as a last resort in refractory VT.	
ADVERSE EFFECTS	Proarrhythmic, especially post-MI (contraindicated). IC is Contraindicated in structural and ischemic heart disease.	

Antiarrhythmics— β -blockers (class II)

	Metoprolol, propranolol, esmolol, atenolol, timolol, carvedilol.
MECHANISM	Decrease SA and AV nodal activity by $\downarrow$ cAMP, $\downarrow$ Ca ²⁺ currents. Suppress abnormal pacemakers by $\downarrow$ slope of phase 4. AV node particularly sensitive— $\uparrow$ PR interval. Esmolol very short acting.
CLINICAL USE	SVT, ventricular rate control for atrial fibrillation and atrial flutter, prevent ventricular arrhythmia post-MI.
ADVERSE EFFECTS	Impotence, exacerbation of COPD and asthma, cardiovascular effects (bradycardia, AV block, HF), CNS effects (sedation, sleep alterations). May mask the signs of hypoglycemia. Metoprolol can cause dyslipidemia. Propranolol can exacerbate vasospasm in vasospastic angina. β -blockers (except the nonselective α - and β -antagonists carvedilol and labetalol) cause unopposed α_1 -agonism if given alone for pheochromocytoma or for cocaine toxicity (unsubstantiated). Treat β -blocker overdose with Glucagon, Atropine, Saline (GAS).



**Antiarrhythmics—
potassium channel
blockers (class III)****A**miodarone, **I**butilide, **D**ofetilide, **S**otalol.**AIDS.**

MECHANISM

↑ AP duration, ↑ ERP, ↑ QT interval.

CLINICAL USE

Atrial fibrillation, atrial flutter; ventricular tachycardia (amiodarone, sotalol).

ADVERSE EFFECTS

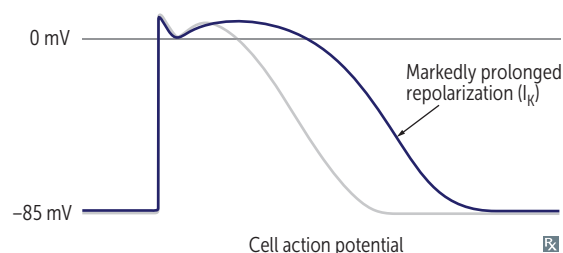
Sotalol—torsades de pointes, excessive β blockade.

Ibutilide—torsades de pointes.

Amiodarone—pulmonary fibrosis, hepatotoxicity, hypothyroidism or hyperthyroidism (amiodarone is 40% iodine by weight), acts as hapten (corneal deposits, blue/gray skin deposits resulting in photodermatitis), neurologic effects, constipation, cardiovascular effects (bradycardia, heart block, HF).

Remember to check PFTs, LFTs, and TFTs when using amiodarone.

Amiodarone is lipophilic and has class I, II, III, and IV effects.

**Antiarrhythmics—
calcium channel
blockers (class IV)****D**iltiazem, **v**erapamil.

MECHANISM

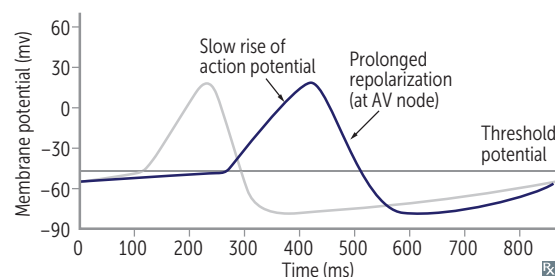
Decrease conduction velocity, ↑ ERP, ↑ PR interval.

CLINICAL USE

Rate control in atrial fibrillation/flutter, prevention of nodal arrhythmias.

ADVERSE EFFECTS

Constipation, gingival hyperplasia, flushing, edema, cardiovascular effects (HF, AV block, sinus node depression).

**Other antiarrhythmics****Adenosine**↑ K^+ out of cells → hyperpolarizing the cell and ↓ I_{Ca} , decreasing AV node conduction. Drug of choice in diagnosing/terminating certain forms of SVT. Very short acting (~ 15 sec). Effects blunted by theophylline and caffeine (both are adenosine receptor antagonists). Adverse effects include flushing, hypotension, chest pain, sense of impending doom, bronchospasm.**Magnesium**

Effective in torsades de pointes and digoxin toxicity.

Ivabradine

MECHANISM

Ivabradine prolongs slow depolarization (phase “IV”) by selectively inhibiting “funny” sodium channels (I_f).

CLINICAL USE

Chronic HFrEF.

ADVERSE EFFECTS

Luminous phenomena/visual brightness, hypertension, **bradycardia**.